BCA SYLLABUS

SEM I

S. No.	Paper Number	Paper Name
1.	101	Information Technology and Applications
2.	102	Discrete Mathematical Structure
3.	103	Digital Electronics
4.	104	Elements of System Analysis and Design
5.	105	Programming & Problem Solving Through 'C'
6.	106	Practical (C & MS Office)

SEM II

S. No.	Paper Number	Paper Name
1.	201	Data Structures Using C
2.	202	Oral and Written Communication Skills
3.	203	Multimedia Applications
4.	204	Programming and Problem Solving through C++
5.	205	Operating System
6	206	Practical (Data Structures, C++)

SEM III

S. No.	Paper Number	Paper Name
1.	301	Data Base Management System (Using SQL Server)
2.	302	Programming in PYTHON
3.	303	Computer Architecture
4.	304	Designing and Analysis of Algorithms
5.	305	Organizational Behavior
6.	306	Practical (DBMS, Python)

SEM IV

S. No.	Paper Number	Paper Name
1.	401	Computer Networks
2.	402	Java Programming
3.	403	Software Engineering
4.	404	Web Designing
5.	405	Computer Oriented Numerical Methods
6.	406	Practical (Java, HTML), Mini Project

SEM V

S. No.	Paper Number	Paper Name
1.	501	Programming and Problem Solving Through C#
2.	502	Mobile Application Development Using Flutter
3.	503	Artificial Intelligence and Data Analytics
4.	504	Software testing and Quality Management
5.	505	Computer Graphics
6.	506	Practical (C#)

SEM VI

S. No.	Paper Number	Paper Name
1.	601	E-Commerce
2.	602	Linux & Shell Programming
3.	603	Cryptography and Network Security
4.	604	Industrial Training and Project

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER I

PAPER NUMBER: 101
PAPER NAME: Information Technology and Applications

UNIT -I

Introduction, Characteristics of Computers, Block diagram of computer
Types of computers and features: Mini Computers, Micro Computers, Mainframe Computers, Super Computers(02).Types of Programming Languages: Machine Languages, Assembly Languages, High Level Languages(02), Translators:- Assembler ,Compiler, Interpreter and Linkers(02),Operating system concepts, Types of OS, Functions of OS(04).
Lectures : 10

UNIT -II

I/O Devices:- Keyboard, Mouse, Scanner, Light pen, Trackball, Joystick, Bar CODE reader, OCR, OMR, MICR, Digitizer; Monitor, Printer, Plotter etc., (03) Memory concepts, Types of Memory (Primary and Secondary):-RAM,ROM and its types, Secondary Storage Devices (Magnetic tape, Magnetic Disk(FD, HD),Optical Disk(CD,DVD), Pen drive) (04).Data Organization: Drives, Files, Directories(03).
Lectures : 10

UNIT -III

Basic elements of a communication system, Data transmission modes, Data Transmission speed, Data transmission media(03),Digital and Analog Transmission, Network topologies, Network Types (LAN, WAN and MAN), Client and Servers , Intranet, Extranet.(03)
Internet: Introduction to Internet, Terminologies related to Internet: Protocol, Domain name, IP address, URL, World Wide Web, Connecting to the Internet,(04)
Lectures : 10

UNIT -IV

Introduction to Big Data: Types of digital data, Big Data architecture and characteristics, Big Data technology components, Big Data importance and applications.(5)
Introduction to Block Chain: Structure, Operational aspects of Bitcoin Block, Compare different types of Block chains, Protocols, Payment mode code execution.(5)
Lectures : 10

Total Lectures:40

Reference Books :

1. Fundamental of Computers – By V. Rajaraman B.P.B. Publications
2. Fundamental of Computers – By P. K. Sinha
3. Michael Minelli, Michelle Chambers, and Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER I

PAPER NUMBER: 102
PAPER NAME: Discrete Mathematical Structures

UNIT -I

Sets: Sets, Subsets, Equal Sets Universal Sets, Finite and Infinite Sets, Operation on Sets, Union, Intersection and Complements of Sets, Cartesian product, Cardinality of Set, Simple Applications.(04)

Relations And Functions**:** Properties of Relations, Equivalence Relation, Partial Order Relation Function: Domain and Range, Onto, Into and One to One Functions, Composite and Inverse Functions, Introduction of Trigonometric, Logarithmic and Exponential Functions.(06)

Lectures : 10

UNIT -II

Partial order relations and lattices: Partial Order Sets, Representation of POSETS using Hasse diagram, Chains, Maximal and Minimal Point, Glb, lub(05), Lattices & Algebraic Systems, Principle of Duality, Basic Properties, Sub lattices, Distributed & Complemented Lattices(05)

Lectures : 10

UNIT -III

Number Theory: Greatest Common Divisors, Euclidean Algorithms, Fibonacci Numbers, Complexity of Euclidean Algorithms, Congruences and Equivalence Relations, Introduction to Public Key Encryption Schemes, Dividends(04)

Groups & Sub groups: Group axioms, permutation groups, subgroups, cosets, normal subgroups, semi-groups, free semi-groups, monoids (04), sequential machines, error correcting codes, modular arithmetic grammars(02).

Lectures : 10

UNIT -IV

Combinatorics & Recurrence Relations: Basic Theorems on permutation and combinations. Pigeon hole principle, principle of inclusion and exclusion(04). Ordinary & exponential generating functions,(02) recurrence relation, solving recurrence relation by substitution, solving recurrence relation by conversion to linear recurrence relation(04)

Lectures : 10

Total Lectures:40

Text Book:

1. Kolman, Busby and Ross, “Discrete Mathematical Structure”, PHI, 1996.

Reference Books:

1. H.K. Dass, “Advanced Engineering Mathematics”; S.Chand & Co., 9th Revised Ed.,2001.
2. S.K. Sarkar, “Discrete Maths”; S. Chand & Co., 2000 Kolman, Busby & Ross "Discrete Mathematical Structures"
3. Trembly. J.P. & Manohar. P "Discrete Mathematical Structures with Applications to computer Science"
4. C.L.Liu, "Elements of Discrete Mathematics"

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER I

PAPER NUMBER: 103
PAPER NAME: Digital Electronics

UNIT -I

Binary System : Digital Computers and digital systems , Number system(03), Binary Arithmetic , Signed binary numbers, complements : r's complements(2's complement, 10's complements), (r-1)'s complements (1's complement, ,9's complement) , Subtraction using 1's complement and 2's complement method , Binary codes ,Logic gates: Inverter, AND, OR, NOR, NAND, XOR, XNOR, De-Morgan's Theorems, Boolean algebra, Canonical and standard forms , Karnaugh Map (04) , Introduction to IC Digital logic families(RTL, DTL, TTL, ECL, MOS and CMOS) and characteristics (Fan-In ,Fan-out, Power Dissipation , Propogation delay ,Noise margin) of IC Digital logic families (TTL , ECL , CMOS) (03)
Lectures: 10

UNIT -II

Combinational and sequential Circuits : Half adder , Full adder , Half Subtractor , full Subtractor , Serial and parallel adder (04) , Code conversion circuits(02) , Parity generator and checker (02) , Comparators(01) , Encoder , Decoder , Multiplexer , De-multiplexer (02)
Lectures: 10

UNIT -III

Flip flops: RS-Flip flop, D-Flip flop, T-flip flop, JK-Flip flop , JK Master slave flip flop, edge triggered and pulse triggered flip flops (05), Registers: Simple Register , Shift registers ,Types of Shift Register , Construction and timing diagram of 4-bit Shift Register (05).
Lectures: 10

UNIT -IV

Counter - Binary counter, characteristic of counter-(Synchronous/Asynchronous, Modulus/length of counter, UP/DOWN counter, Speed of Asynchronous counter) Construction of full length and Non full length, UP/DOWN Synchronous counter using 2, 3 and 4 FF, Ripple counter. Design of different MOD counter, Construction of Asynchronous counter, Ring Counter, Johnson Counter(05) .
Random Access Memory, Read Only Memory, Types of Read Only Memory(02), Multivibrators- Astable, Monostable, Bistable, Schmitt Trigger, Timer (03)
Lectures: 10

Total Lectures: 40

Reference Books:

1. Digital Electronics- By Morris Mano
2. Electronics- V.K. Mehta
3. Digital Electronics- R.P. Jain

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER I

PAPER NUMBER: 104
PAPER NAME: Elements of System Analysis and Design

UNIT -I

System concept, Definition, Characteristics and Elements of System,(04) **Types of System:**-Abstract and Physical system, Open and Closed System, Deterministic and Probabilistic System, Man made Information System, System Models and types of models, System environment and boundaries,(04) **system analyst** ,role of system analyst, qualification and responsibilities of System Analyst(02).
Lectures:10

UNIT -II

System development life cycle and its various phases:-Preliminary investigation, determination of system requirements, Development of software, System testing, Implementation, evaluation and maintenances,(05) System Flow chart and its symbols **Software Crisis**: From programmers' point of view, from users' point of view. (05)
Lectures:10

UNIT -III

System Analysis:-System Planning, Information Gathering and its Tools,(02) Feasibility study ,steps in feasibility analysis , its report and importance,(03) various tools of Structured Analysis:- data flow diagrams, Data Dictionary, Decision Tree, Structured English, Decision Table(02),Cost /Benefit Analysis, Introduction of Control Flow Graph, CPM, PERT chart, Gantt chart.(03)
Lectures:10

UNIT -IV

System design:-process modeling, logical and physical design, design methodologies(03) **data base design**:-objectives of database,Types of relationship, types of data structure, (03)**system testing and quality assurance**:-reason of system testing,nature of test data,Test plan,types of System test ,Quality assurance goals in SDLC,levels of Quality Assurance(02) ,**implementation and software maintenance**:-Introduction of implementation,primary activities of a maintenance procedure and reducing maintenance cost(02).
Lectures:10

Total Lectures:40

Text Books :

1. System Analysis and Design by Elias M. Awad.
2. Software Engineering by Pressmen.
3. System analysis and design By Kendell & kendell

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER I

PAPER NUMBER: 105
PAPER NAME: Programming and Problem Solving Through 'C' Language

UNIT-I
Fundamentals of C programming, Programming Techniques and Concepts, Overview of C, History and Structure(02), C character set, Identifiers and Keywords, Data types, Data Types Declarations, Constants and Variables, Expression and statement and symbolic constants, Basic I/O(03), Preprocessor command: #include, define, preparing and running a complete C program.(03)
Lectures : 10

UNIT-II
Operator and expression: Arithmetic, unary, logical, bitwise, assignment and conditional operators, library functions, (04)Construction of loops and implementation, control statement: (03)While, Do-While, For Statements nested loops, If-else, switch, break, continue and go-to statements, comma operator. (03)
Lectures : 10

UNIT-III
Array one dimensional and two dimensional arrays, Null terminated Strings as Array of Characters,(03) Functions, functions prototypes, function call, call by value, call by reference, Recursion(03), storage classes: automatic, external and static variables(03),Pointer: Declaration, uses of pointers, array of pointers, Passing pointer to a Function, Operations on Pointers. (04)
Lectures : 10

UNIT-IV
Structure variables, Initialization, Structure Assignment, Nested Structures, Structures and Functions, Structures and Arrays. Unions Declarations,(06) File handling: Open, Close, Create, Process, unformatted data file.(04)
Lectures : 10

Total Lectures: 40

Reference Books :
1. Fundamental of Computers – By V. Rajaraman B.P.B. Publications
2. Fundamental of Computers – By P. K. Sinha
3. C in Depth -S.K.Srivastava,Deepali Srivastava
4. Let us C -Yashvant Kaniktar

PAPER NUMBER: 106
PAPER NAME: Mini Project Using C

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER II

PAPER NUMBER: 201
PAPER NAME: Data Structure Using C

UNIT-I
Data Structure, definition, and application, Arrays: Representation of single and multidimensional arrays; sparse arrays - lower and upper Triangular matrices and Tri-diagonal matrices, Recursion (01), Defining Stack and Queue, Stack Operations and Implementation, Array Implementation, Pointer Implementation, Stack Applications, Convert Number Bases by Using Stacks, Infix to Postfix Conversion, Evaluation of postfix expression.(03) Introduction and primitive operation on queues, D-queues Queue Application, Circular, Double Ended and Priority Queues.(03)
Lectures: 10

UNIT-II
Lists, Basic Terminology, Static Implementation of Lists, Pointer Implementation of Lists, Insertion in a list, Deletion from a list, traversal, Searching,(03) Arrays using Linked List, Doubly Linked Lists, Circular, Doubly, Circular Doubly Linked List(07)
Lectures: 10

UNIT-III
Defining Graph, Basic Terminology, Graph Traversal: Depth First Search (DFS), Breadth First Search (BFS), Shortest Path Problem,(03) Trees: Introduction and terminology; Traversal of binary trees; Algorithms for tree operations such as traversal, insertion, deletion; Binary Trees, Inorder, Postorder and Preorder Traversals,(03) Minimal Spanning Tree, Prims and Kruskals Algorithm, Binary Search Trees, Operations on a BST, Insertion, Deletion , Search for a key in BST, B-Trees: Introduction: Threaded Binary tree, AVL Trees(04)
Lectures: 10

UNIT-IV
Searching and Sorting techniques, Sequential Search, Binary Search, Selection Sort, Insertion Sort, Bubble Sort(05), Quick Sort, 2-way Merge Sort, Heap Sort, Bucket or Radix sort. Hashing.(05)
Lectures: 10

Total lectures:40

Reference Books:

1. Data Structure through C- G.S.Baluja
2. Data Structures Using C: Aaron M. Tannenbaum

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER II

PAPER NUMBER: 202
PAPER NAME: Oral and Written Communication Skills

UNIT-I: Introduction to Communication : Meaning and process of communication. Scope of commu nication (02 lectures) Channels of communication. (01 lecture) Models of communication – SMCR Model, Shannon-Weaver Model (03 lectures) Barriers to communication (01 lecture) Verbal communication, 7 C's of communication (02 lectures) Meta Communication (01 lecture)

Lectures: 10

UNIT-II: English Grammar and Written Communication English Grammar – Tense/Article, Preposition, Modals, Transformation of sentences. (04 lectures) Written communication – principles, merits & demerits. (01 lecture) Business letters – Structure/Parts of a letter, essentials of effective business letters. (02 lectures) Sales letters – qualities, using effective sales statements (01 lectures) Types of letters – sales, complaints and adjustments, placing orders (02 lectures)

Lectures: 10

UNIT-III: Oral Communication Oral communication – advantages and disadvantages. (01 lecture)Interviews – meaning and purpose, Types of interview. (02 lectures) Guidelines for interviewer and interviewee. (02 lectures) Important non-verbal aspects in an interview - Proxemics, Paralanguage. (02 lectures) Styles of interviewing, Cover letter, Job application, Resume-writing. (03 lectures)

Lectures: 10

UNIT-IV: Reading and Listening Reading – Meaning and importance in communication (01 lecture)Listening – meaning, nature and importance, Barriers to listening (03 lectures) Kinesics – Role of body language in communication (02 lectures) Group discussion – methodology, guidelines, role and functions (02 lectures) Practice role playing and group discussions (02 lectures)

Lectures: 10

Total Lectures: 40

Suggested Books:
"Business Communication" By Meenakshi Raman & Prakash Singh
"Business Communication" By P D Chaturvedi & Mukesh Chaturvedi
"Essentials of Business Communication" By Rajinder Pal & J S Korlahalli
"Business Communication" By Namita Gopal

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER II

PAPER NUMBER: 203

PAPER NAME: Multimedia Applications

UNIT-I

Introduction to Multimedia, Multimedia Information, Multimedia Objects (03). Multimedia in Business, Communication, Edutainment, Entertainment, Public Access and Knowledge Transfer (04); Recent developments in Multimedia. (03)

Lectures: 10

UNIT-II

Multimedia Project and its Stages: Planning, Production and Distribution. (03) Multimedia Hardware, Memory & Storage devices, Communication devices, Multimedia Softwares, Presentation tools, Tools for Object Generation, Video, Sound, Image Capturing, Authoring tools (04). Multimedia Building Blocks: Text, Audio, Video, Graphics and Various file formats (03).

Lectures: 10

UNIT-III

Data Compression: Huffman Coding and Algorithm, Shannon Fano Algorithm, Adaptive Coding, Arithmetic Coding, Higher Order Modeling. Finite Context Modeling (04), Dictionary based Compression, Sliding Window Compression, Compression Ratio, Lossless & Lossy Compression, Speech Compression & Synthesis (03). Digital Audio concepts, Sampling Variables, Silence Compression. (03)

Lectures: 10

UNIT-IV

Images, Bitmaps, Vector drawing, Image File, Image Standards, JPEG Compression, Zig Zag Coding (04). **Video**: Video representation, MPEG standards, MHEG Standard Video Streaming on Net (04), Video Conferencing, Multimedia Broadcast Services (02).

Lectures: 10

Total Lectures: 10

Reference Books:

1. Tay Vaughan “Multimedia, Making IT Work” Osborne McGraw Hill.
2. Buford “Multimedia Systems” Addison Wesley.
3. Mark Nelson “Data Compression Book” BPB.
4. David Hillman “Multimedia technology and Applications” Galgotia publications.
5. Rosch “Multimedia Bible” Sams Publishing.
6. Sleinreitz “Multimedia System” Addison Wesley.

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER II

PAPER NUMBER: 204
PAPER NAME: Programming & Problem Solving Through C++

UNIT-I
OOP concept, Procedural vs OOP Programming, OOP terminology and features (03), Tokens, Character set, Keywords, Data-types Declarations, Constants and variables, Expressions , Standard Library and header files(03) Operator and Expressions; Arithmetic Operator, Increment/Decrement Operator, Relational Operator, Logical Operator and Conditional Operators, Logical Expression, Flow of control statement; Selection statement, Iteration Statement, Jump statement, Construction of loops and implementation , while, Do-while, for statements nested loops.(04).
Lectures:10

UNIT-II
Arrays one dimensional and two arrays.(01)
Classes and Objects: Need for Classes, Encapsulation, Information Hiding (03), Declaration of Classes, iostream operators, referencing class Members,(03) Scope of Class and Its members, Scope resolution operator and its uses, Nested Classes(03)
Lectures:10

UNIT-III
Functions in class: function definition, Default arguments, Constants arguments, Call by value, Call by reference, Calling Functions with arrays, returning from a function,(03) storage class specifies automatic, external and static variable(02), Function overriding, Function overloading, Operator Overloading. Functions: Friend, Inline, Abstract, Virtual, Pure Virtual.(05)
Lectures:10

UNIT-IV
Constructors and Destructor: Declaration, Definition and characteristics(02) Default Constructor, Copy constructor, Parameterized constructor, constructor overloading (04) Inheritance: Need, Different forms, single Inheritance, Multiple Inheritance.(04)
Lectures: 10

Total Lectures: 40

Text Books:
1. Introduction to C++, E. Balaguruswamy
2. Let us C++, Yashavant Kanetkar

Reference Book:
1. C++, Sumita Arora

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER II

PAPER NUMBER: 205
PAPER NAME: Operating System

UNIT-I
Introduction to OS,(03) Function of OS-Process Management, Device Management, Memory Management, Information Management,(03) Types of an operating system: Batch System, Multiprogramming and Timesharing ,Parallel ,Distributed and real time System(04)
Lectures : 10

UNIT-II
Process Management: Process Concept, Process Scheduling, CPU Scheduling Criteria, Preemptive and Non Preemptive Scheduling, Scheduling Algorithm: FCFS, SJF, RR (05)
Dead Lock: Resource allocation graph, Introduction: Characterization, Deadlock conditions, Dead lock prevention, Avoidance and Detection. Safe state condition. (05)
Lectures : 10

UNIT-III
Memory Management: Logical and Physical address space, Swapping, Contiguous allocation space, Paging, Fragmentation: Internal and External (05) Virtual Memory Management: Demand paging, Page replacement algorithm: FIFO, LRU, OPR(05)
Lectures : 10

UNIT-IV
File System: File concepts, Access methods, Directory Structure File-System Implementation: File system structures, Directory Implementation, Allocation methods(05)
Disk Management: Disk Structure & Scheduling Methods, Disk management FCFS, SSTF, SCAN, LOOK, C-SCAN, C-LOOK (05)
Lectures : 10

Total Lectures:40

Text Books:
1. Silbersachatz and Galvin, “Operating System Concepts”, Pearson, 5th Ed., 2001
2. Madnick E., Donovan J., “Operating Systems”, Tata McGraw Hill, 2001

Reference Books:
1. Tannenbaum, “Operating Systems”, PHI, 4th Edition, 2000
2. William Stallings “Operating System Concepts”

SEMESTER II

PAPER NUMBER: 206
PAPER NAME: Practical(Data Structure, C++)

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER III

PAPER NUMBER: 301
PAPER NAME: Database Management System (Using SQL Server)

UNIT -I
Data, Information and knowledge, introducing database and different kinds of database users, concept or a database, interacting with a database,(03) architecture of a database, using relational databases, basics of relational databases, using relational databases(03), identifiers for relations, characteristic of database, database system concepts and data independence, content of data dictionary, data administration function.(04)
Lectures: 10

UNIT -II
Traditional data model – ANSI/SPRC, 3-level architecture, over view, of three traditional models – hierarchical, network and relational models, comparison of these models(02), ER model(02), File organization technique – random, file organization technique, multi key file organization technique, Indexing (02), concurrency control, database security, database recovery(04)
Lectures:10

UNIT -III
Data Normalization (02) Introduction to SQL: Characteristics and advantages, SQL Data Types and Literals, DDL, DML, SQL Operators (04), Tables: Creating, Modifying, Deleting, Views: Creating, Dropping, Updating using Views SQL DML Queries: SELECT Query and clauses, Set Operations, Predicates and Joins,.Aggregate Functions, Nested Queries(04)
Lectures:10

UNIT -IV
Specifying constraints and indexes in SQL, data manipulation, multiple table operations (05)
Database Modification using SQL Insert, Update and Delete, T-SQL: Cursor, concept of Stored Procedures, Functions and Triggers (05)
Total Lectures:10

Total Lectures:40

Text Books:
1 R. Elmarsi and SB Navathe, “Fundamentals of Database Systems”, Addison Wesley,4th Ed., 2004

Reference Books:
1. Abraham Silberschatz, Henry Korth, S. Sudarshan, “Database Systems Concepts”, 4th Edition, McGraw Hill, 1997.
2. Jim Melton, Alan Simon, “Understanding the new SQL: A complete Guide”, Morgan Kaufmann Publishers, 1993.
3. A. K. Majumdar, P. Battacharya, “Data Base Management Systems’, TMH, 1996.4. Bipin Desai, “An Introduction to database Systems”, Galgotia Publications, 1991.

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER III

PAPER NUMBER: 302
PAPER NAME: Programming in Python

UNIT I:
Introduction to Python: Features, Installation, Setting up path and Working with Python Understanding Python variables Python basic Operators Understanding python blocks, Python Data Types: Declaring and using Numeric data types: int, float, complex Using string data type and string operations Defining list and list slicing Use of Tuple data type. Python Program Flow Control Conditional blocks using if, else and nested if-else Simple for loops in python For loop using ranges, string, list and dictionaries Use of while loops in python, Nested loop, Loop manipulation using pass, continue, break and else.
(8 Lectures)

UNIT II:
String Manipulation: Accessing Strings, Basic Operations, String slices. Lists: Introduction, Accessing list, Operations, working with lists. Function and methods, Tuples: Accessing Tuples, Operations, working, Function and methods. Dictionaries: Accessing values in Dictionaries, working with dictionaries, properties, functions.
(7 Lectures)

UNIT III:
Functions: Defining a function, calling a function, Types of functions, function arguments, Anonymous functions, Global and local variables. Modules: Importing modules, Math module, Random module, Packages, Composition.
(7 Lectures)

UNIT IV:
Input Output: Printing on screen, reading data from keyboard, opening and closing files, reading and writing files, functions. Exception Handling. OOPs concept: Class and object, Attributes, Inheritance, Overloading, overriding, Data hiding, Regular Expressions.
(8 Lectures)

Books:

1. Python programming for Beginners by Michael Knapp
2. Core Python Programming : Dreamtech.
3. Learning Python by O Relly

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER III

PAPER NUMBER: 303
PAPER NAME: Computer Organization and Architecture

UNIT-I
Register Transfer Language , Register Transfer, Bus and memory Transfer,(03) Micro-Operations, Arithmetic Micro-Operation, Logic Micro-Operation, Shift Micro-Operation, Common Bus Organization ,Multiple Bus Organization, Arithmetic Logic Shift UNIT (03), Addition and Subtraction Algorithm(01) ,Multiplication Algorithm (Booth Multiplication Algorithm) (03)
Lectures: 10

UNIT-II
Instruction Codes , Computer Instruction(02) , Computer Registers (DR, AR, AC, IR, PC, TR, INPR, OUTR) , Instruction Cycle (Fetch and Decode) (03), **Control UNIT Organization**: Functional Requirements of a Control UNIT, Hardwired Control UNIT(02) , Micro programmed Control UNIT (Microinstructions , Micro program Sequencer) (03)
Lectures:10

UNIT-III
Processor Design: General Register Organization , Stack Organization(Push and Pop Operation , Register Stack , Memory Stack) (02), Instruction Format(Zero Address Instructions , One Address Instruction ,Two address Instruction , Three Address Instruction) ,Data transfer and manipulation(Data transfer Instructions , Data Manipulation Instructions , Program Control Instructions) (03) ,RISC and CISC Architecture (02) , Addressing Schemes: Immediate Addressing, Direct Addressing, Indirect Addressing, Register Addressing, Register Indirect Addressing, Displacement Addressing (03)
Lectures: 10

UNIT-IV
Input-Output Organization: I/O Interface, Modes of transfer, Interrupts & Interrupt handling, Direct Memory access (DMA controller , DMA transfer)(04) ,**Memory Organization:** Memory Hierarchy, Main Memory (RAM and ROM Chips)(02), Auxiliary memory(Magnetic disks , magnetic Tapes), Cache memory (Hit Ratio , mapping techniques :Associative mapping ,Direct mapping ,Set associative mapping), Introduction to Virtual Memory(04).
Lectures:10

Total Lectures:40

Text Book:
1. Computer System Architecture, M. Mano(PHI)

Reference Books:
1. Computer Organization, Vravice, Zaky & Hamacher (TMH Publication)
2. Structured Computer Organization, Tannenbaum(PHI)
3. Computer Organization, Stallings(PHI)
4. Computer Organization, John P.Hayes (McGraw Hill)

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER III

PAPER NUMBER: 304
PAPER NAME: **Designing and Analysis of Algorithms**

UNIT – I
Mathematical Preliminaries: Review of growth functions, (03 Lectures) Introduction of algorithm, characteristics of algorithm, (03) Complexity of algorithm, Efficiency of Algorithm(02), Asymptotic notations.(02)
Lectures:10

UNIT – II
Sorting and Searching- Insertion sort, Bubble sort, Selection sort, (04) Quick sort, Merge sort, (02) Radix Sort, Bucket Sort, (02 Lectures) External Sorting (01)Searching-Sequential and Binary search and their complexities. (02)
Lectures:10

UNIT -III
Greedy algorithms, general characteristics of greedy algorithms, (03) job sequencing, minimum spanning tree, Single source shortest paths. (03) The knapsack problem, task scheduling problem, (01) Divide and Conquer Technique: merge sort, quick sort, strassen's matrix multiplication (03)
Lectures:10

UNIT -IV
Dynamic Programming: General method Matrix multiplications,Travelling sales person problem, binary search trees, 0/1 knapsack problem(04)

Backtracking – n-Queen's Problem, Hamiltonian Circuit problem, graph colouring (03)

Branch and bound –Traveling salesman problem. FIFO branch and bound(03)
Lectures:10

Total Lectures:40

Text Books:

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, Clifford Stein, "Introduction to Algorithms", 2nd Ed., PHI, 2004.
2. Anany Levitin, "Introduction to the Design and Analysis of Algorithm", Pearson Education Asia, 2003.

Reference Books:

1. A. V. Aho, J. E. Hopcroft, J. D. Ullman, "The Design and Analysis of Computer Algorithms", Addition Wesley, 1998.
2. Ellis Horowitz and Sartaz Sahani, "Computer Algorithms", Galgotia Publications,1999.
3. D. E. Knuth, "The Art of Computer Programming", 2nd Ed., Addison Wesley, 1998

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER III

PAPER NUMBER: 305
PAPER NAME: ORGANIZATIONAL BEHAVIOR

UNIT -I
Introduction to organizational behavior: nature, scope, challenges of organizational behavior, managing people and organization, organizational behavior in global context organizational behavior model.

Lectures:10

UNIT II
Work teams - concept and types. Team effectiveness. Team creation. Power-bases of power, contingency approach to power .Politics -concept, causes and management, leadershipconcept, styles, theories- Trait theory, charismatic leadership theory, behavioral theory

Lectures:10

UNIT III
Organization theory - classical, neoclassical and modern. Organizational culture conceptimpact, creating organization culture. Organizational design-importance types of OD, consequences of poor OD

Lectures:10

UNIT IV
Organizational change-factors, planned change, change resistance to change and overcoming resistance, Organization development- definition, process of organizational development. Action research model, Organizational development interventions-sensitivity training and organizational development

Lectures:10
Total Lectures:40

Suggested readings:
Organizational behavior - Fred Luthans
Organizational behavior - Stephen Robins
Organizational behavior - L.M. Prasad

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER III

PAPER NUMBER: 306
PAPER NAME: Practical (DBMS, Python)

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER IV

PAPER NUMBER: 401
PAPER NAME: Computer Networks

UNIT - I
Computer networks, Networks Hardware-----Local Area network, Metropolitan Area network, Wide Area network, (03) Wireless network, Internetworks, (03) Networks Software: Protocol Hierarchies, Design and Issues for layers, (02) Interfaces and Services, Connection oriented and Connection less Services (02)
Lectures: 10

UNIT- II
Reference Models, and OSI Reference Model, TCP/IP Protocol Suit (03) The Physical Layer: Maximum Data Rate of a channel, Transmission Media: Magnetic Media, Twisted Pair, Baseband and Broadband Coaxial cable, Fiber Optical (03) Wire less Transmission, structure of telephone system, Switching, Multiplexing, ISDN (04)
Lectures: 10

UNIT -III
Data link layer, Error control, Flow control, Sliding Window Protocol, (03) Channel Allocation Problem, Multiple Access Protocol: ALOHA, CSMA protocol, Collision Free protocol, Polling, (04) Network layer: Routing Algorithm, Congestion Control Algorithm, IP protocol, IP Addresses. (03)
Lectures: 10

UNIT -IV
Transport Layer: Addressing, Establishing and releasing a connection, (03) TCP service Model, TCP protocol (04), the Application Layer: Network Security, Domain Name System(03)
Lectures: 10

Total Lectures: 40

Text Book:
1. Computer Networks by Andrew S Tanenbaum

References Books:
1. Computer Netwoks and Internet by Douglas E Comer
2. Data Communication and Networking by Brijender Singh
3. Computer Networks by Peter Norton
4. Computer Networking with Internet Protocols by William Stallings

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER IV

PAPER NUMBER: 402
PAPER NAME: Java Programming

UNIT -I

Java programming language overview, History of Java, features of Java, Java architecture,(04) referring to applets and applications, compiling and running a program, Byte code and JVM, (03) Java primitives data types, Variables: Declaration, Initialization, scope and lifetime, Command line arguments. (03)
Lectures: 10

UNIT -II

Java Operators, Typecasting, (02) Control statements and looping structure: if statement, switch construct, while loop, the for loop, the do loop, the break statement, the continue statement, retun statements, arrays. (02) Classes and object: concept of OOPS, constructors, types of constructors ,garbage collector, finalize(),(03) java method and object declaration, java methods, types of methods, passing arguments, methods overloading, constructor overloading, access specifiers. (03)
Lectures: 10

UNIT -III

Inheritance, using inheritance,this and super keywords, (03) overriding methods, Abstract classes, Package and Interface: defining packages, importing packages, grouping classes in packages, creating Interface, implementing interface, using interface ,extending interface. (04)
String handling: using the string class as data type, using strings, string constructors, using string without the new modifier, using string methods, StringBuffer class.
Vector, exception handling, multithreading (03)
Lectures: 10

UNIT -IV

Graphical user interface: defining applet, applet lifecycle, applet tags, applet methods, drawing lines, rectangles, polygons (03)
AWT package class hierarchy, AWT controls, button, labels, text field, text area, check box, radio box, list, scroll bars, choice, frame, adding a button, creating panels ,layout managers. (04) Introduction to Servlets and RMI(03).
Lectures: 10

Total Lectures: 40

Text Books:

1. J2EE- Balaguruswamy
2. Java- R. Krishnamothy

References Books:

1. Java Black book-Seven Holzner
2. Java: a beginner's guide- Herbert Schildt
3. Core Java-Gary Cornell

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER IV

PAPER NUMBER: 403
PAPER NAME: Software Engineering

UNIT -I

Software Engineering Fundamentals**: Definition** of software product and process(03), Software Crisis, Software development paradigms(03), Software lifecycle models: Waterfall Model, Prototyping Model, Iterative Enhancement Model, Evolutionary Development Model and Spiral Model(04).
Lectures: 10

UNIT -II

Software Requirement Analysis & Specification: System specification, Software requirements specification (SRS) standards (02), Formal specification methods, Specification tools, Requirements validation and management (02).
Software Process: Software Process and Models, Tools and Techniques of Process Modeling, Product and Process(03). important qualities of software product and process: correctness, reliability, robustness, user friendliness, verifiability, maintainability, reusability, portability, data abstraction, modularity, Principles of software engineering(03).
Lectures: 10

UNIT -III

Software Design: Software architecture, Modular design - cohesion and coupling(02), Process-oriented design, Data-oriented design, User-interface design, Real-time software design(03).
CASE Tools: Relevance of CASE tools, High-end and low-end CASE tools(02), Automated support for data dictionaries, DFD, ER diagrams(03).
Lectures: 10

UNIT -IV

Coding and Testing: Choice of Programming languages, Coding standards(02), Introduction to Testing Process, Functional & Structural Testing(02), Testing Activities like UNIT, Integration & System Testing(03), Testing tools and workbenches(03).
Lectures: 10

Total Lectures: 40

Text Book:

1. “Software Engineering-A Practitioner’s approach”-R.S. Pressman

Reference Book:

1. “Software Engineering ” by Pankaj Jalote

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER IV

PAPER NUMBER: 404

PAPER NAME: Web Designing

UNIT -I

Internet evolution of internet, internet application, client and servers, hosts and nodes, internet services, (02) different types of connections, dial up, leased, VSAT, (02) internet service providers, choosing an ISP, (03) DNS system, E-mail, basics e-mail functions, E-mail Protocols. (03)

Lectures: 10

UNIT -II

Introduction *to* HTML-- a short history of World Wide Web ,creating first HTML document, understanding basics of HTML tools, entering tags and attributes, (02) applying structures tags, linking documents, URL, types of URL's, (03) constructing link anchors, inserting e-mail, links, including images, developing images, adding images(03), using images as links, using background images(02)

Lectures: 10

UNIT III

Tables, creating basic tables, spanning rows and columns, adding captions, formatting tables, adding and formatting borders(03).HTML forms, developing forms, creating forms, (04) Frames:creating frames, accommodating non-framed browsers(03)

Lectures: 10

UNIT IV

Java Script: adding java script, what is java script, (02) adding event handlers, Loops, control statements, functions,(03) validations, (02) Introduction to DHTML.Style Sheets. (03)

Lectures: 10

Total Lectures: 40

Text Books:

1. Web Design Complete Reference by Thomas A. Powell
2. JavaScript: The Complete Reference by Thomas A. Powell

Reference Books:

1. HTML Black Book: The Programmer's Complete HTML Reference Book by Steven Holzner
2. HTML:A Beginner's Guide, Second Edition by Wendy Willard
3. Learning Web Design: A Beginner's Guide to HTML, Graphics, and Beyond by Jennifer Niederst

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER IV

PAPER NUMBER: 405
PAPER NAME: Computer Oriented Numerical Methods

UNIT -I
Numerical system and error and types of error, Floating point Arithmetic, Source of error, Zeros of transcendental equations and polynomials (03), system of non linear equation(03), Solution of Algebraic and transcendental function- Bisection Method , Iteration Method, Method of false position, Newton Raphson method, Generalized Newton's method (04).
Lectures: 10

UNIT -II
Solution of system of linear equation, Gaussian elimination method, Gauss Jordon method (03), Pivoting, Iterative methods of Jacobi and Gauss Seidel Methods (03), Matrix Inversion Method (02), Method of Factorization (02).
Lectures: 10

UNIT -III
Interpolation, Errors in Polynomial Interpolation, Finite Differences, Forward differences, Backward Differences, Central Differences (02), Newton's formula for interpolation, Guass's central difference formula (02), Stirling's Formula, Bessel's formula (03), Lagrange's Interpolation Formula, Error in Lagrange's Interpolation Formula (03).
Lectures: 10

UNIT -IV
Numerical differentiation, Error in Numerical differentiation, Cubic Spline method (03), Numerical Integration: Trapezoidal rule, Simpson's 1/3 rule, Simpson's 3/8 rule , Romberg's Interpolation (04), Numerical solutions of ordinary differential equations: Solution by Taylor's series, Euler's Method, Runge-kutta Methods (03).
Lectures: 10

Total Lectures: 40

Text Book:
1. Computer Based Numerical & Statistical Techniques – by Manish Goyal

Reference Books:
1. Numerical Analysis –by Goel and Mittal
2. Numerical Methods in Engineering and Science – by Dr. B. S. Grewal
3. Numerical Methods – by S. S. Sastry

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER IV

PAPER NUMBER: 406
PAPER NAME: Practical (Java, HTML), Mini Project

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER V

PAPER NUMBER: 501
PAPER NAME: Programming and Problem Solving through C#

UNIT -I
The .net Framework: introduction, common language runtime, common type system, common language specification, (03) the base class library, the .net class library, Intermediate language, (03) Just in time compilation, garbage collection, assemblies, web services, COM, localization(04)
Lectures: 10

UNIT -II
C# Basics: Introduction, data types, identifiers, variables, constants, C# statements, OOPs concept, (03) array and strings, operators, control statements, classes and objects(03), access modifiers, overloading, inheritance, overriding, interfaces. (04)
Lectures: 10

UNIT -III
Visual studio IDE features, introduction to Window forms, components, (02) control: textbox, label, linklabel, status bar, (02) checkedlistbox, combobox, listbox, listview, radiobutton, button, panel, (03) groupbox, dialog box, menu control, properties, methods, events of controls. (03)
Lectures: 10

UNIT -IV
ADO.net, the component model, creating database connection, database command, (02) data repeater, connecting to data sources, choosing a .net data provider, (02) manage a connection, building command objects, (03) executing commands, building datasets and datatables, data adapter (03)
Lectures: 10

Total Lectures: 40

Text Book:
1. C# Programming for beginners By Mahesh Chand

Reference Books:
1. The Complete Visual C# Programmer's Guide
2. A Programmer's Introduction to C# 2.0, Third Edition
3. C# and the .NET Platform, Second Edition

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER V

PAPER NUMBER: 502
PAPER NAME: MOBILE APPLICATION DEVELOPMENT USING FLUTTER

Unit-I
Dart programming Basic: features, installation, basic syntax, comments ,keywords, data types, variables , Operators, **Control flow Statement:** if statement, if else, if elseif, switch, for,for..in, while,do while Loop.
Function: Dart function,Anonymous Function,main function ,Recursion
Lectures : 10

Unit-II
Object Oriented concepts:-Class & object, constructor,this keyword,static keyword,super keyword,inheritance,super constructor,methods,overriding,getter and setters, Abstract class,Interface
Dart Advance: Exception handling,typedef,metadata,collection,generics,packages,libraries,generators
Lectures : 10

Unit-III
Flutter basics: Installation, Architecture, widgets, Layout,gestures,state management ,flutter IDE
Flutter widgets: Scaffold,container,row and column,Text,TextField,Buttons,Stack,Forms,Alert Dialog,icons,images,card,Tabbar,Drawer,Lists,Gridview,Toast,checkbox,radio button,Progressbar,Slider,Navigation Bar
Lectures : 10

Unit-IV
Android platform,packages,splash screen google maps,REST API,Database,Testing
Lectures : 10

Total Lectures: 40

Reading:

- Beginning App Development with flutter by Rap Payne ,Apress publication
- Beginning Flutter: A Hands on Guide to App development
- Flutter Complete Reference by Alberto Miola

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER V

PAPER NUMBER: 503
PAPER NAME: Artificial Intelligence and Data Analytics

UNIT -I
Introduction, The importance of AI , Applications of AI , general issue in AI problem solving (02), production system (02), knowledge: Definition & importance, knowledge based systems , components of knowledge based system (03), intelligent agents (03).
Lectures: 10

UNIT -II
Search and Control Starategies, Examples of Search Problems(The eight puzzle, Travelling Salesman Problem) (03), Uninformed Search (Breadth First Search , Depth First Search , Depth First Iterative Depending Search , Bidirectional Search) and informed search (Heuristic Information , Hill Climbing Methods ,Best First Search , Branch and Bound Search ,A* Search) (04), Searching AND-OR Graphs, The AO* Algorithm , Game Playing : mini max search approach (03).
Lectures: 10

UNIT -III
Formalized Symbolic logics: Propositional logic , Syntax and Semantics for Propositional Logic ,Limitation of Propositional Logic (02), First order predicate Logic , Syntax and Semantics for FOPL ,Properties of Well Formed Formula (02), skolemisation, Conversion to Clausal Form (02) , inference rules (02), unification, resolution principle, (02).
Lectures: 10

UNIT -IV
Semantic networks, frame system(03), value inheritance scripts(02), LISP and other AI programming languages(03), Introduction to Data Analytics: Sources and nature of data, classification of data (structured, semi-structured, unstructured), characteristics of data, introduction to Big Data platform, need of data analytics, evolution of analytic scalability, analytic process and tools, analysis vs reporting, modern data analytic tools, applications of data analytics.(02).
Lectures: 10

Total Lectures: 40

Text Book:
1. D.W. Patterson, "Introduction to AI and Expert Systems", PHI
2. E. Rich and K. Knight, "Artificial intelligence", TMH

Reference Book:
1. Nils J Nilsson ,"Artificial Intelligence -A new Synthesis" Harcourt Asia Ltd.

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER V

PAPER NUMBER: 504
PAPER NAME: Software Testing and Quality Management

UNIT -I

Testing Software :Software Testing and its importance, Module Testing, Integration Testing, Top-down versus bottom up testing, Mixed testing.(03)

Software Faults and Failures: Types of faults, Origins of Software Defects, The Cost of Repairing Defects(02)Verification and Validation :Verification Testing, Coding standards, Walk-Through, Formal Inspection(02), Verifying . Requirements, Verifying Functional Design, Validation Test Criteria, Design metrics. (03)

Lectures: 10

UNIT -II

Testing Techniques and Strategies :White-Box Testing: Flow graph notation, Cyclomatic Complexity, Control Structure and Loop Testing, Dataflow Testing.(03)

Black-Box Testing: Graph-based testing methods, Equivalence partitioning, Boundary Value Analysis(03),UNIT Testing, Integration Testing, System Testing, Exhaustive Testing.(04)

Lectures: 10

UNIT -III

BuildingTests and Test Plans :Designing and Creating Tests, Maintaining Checklists, White-box Test Cases and Test Procedures, Test Data Selection and Outputs, Black-box test cases and test procedures, Planning and Creating Test Plans. (04)

Testing Specialized Systems and Applications :Graphical User Interface (GUI) Testing, Usability Testing, Client/Server Architectures and Web Testing, Testing OO Systems, Volume and Stress Testing (03)

Testing Measurements: Software Size and Complexity, Function Point Analysis(03)

Lectures: 10

UNIT -IV

Quality Assurance and Standards:Quality and Quality Assurance (QA), Techniques of quality Assurance, Software Testing and QA,(04)Software Development Models, Quality metrics(02), Configuration Management, Quality management models *(ISO,* SPICE, IEEE, and CMM).(04)

Lectures: 10

Total Lectures: 40

Text Books:

1. W.M. Perry, “Effective Methods for Software Testing", 2002, Wiley.
2. Nina Godbole, “Software Testing and Quality Management”
3. .A.Behforooz and F.Hudson, “Software Engineering Fundamentals", 1996, OUP.

Reference Books:

1. James Peters & W. Pedrycz, "Software Engineering", 2000, Wiley.
2. S. Pfleeger, "Software Engineering", 2001 , Peterson Education.
3. A.Behforooz and F.Hudson, “Software Engineering Fundamentals", 1996, OUP.

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER V

PAPER NUMBER: 505
PAPER NAME: Computer Graphics

UNIT -I

Application of computer graphics, graphics devices- LED,LCD,CDA,FPD (04), drawing geometry, Line drawing Algorithms: DDA and Bresenham, Functions implementation(03), Cathode Ray Tube implementation.(03)
Lectures: 10

UNIT -II

2D transformation: Translation, Rotation, Scaling, Reflection, Shearing(02), Circle Drawing: Bresenham and Mid-point sub divison function implementation(02), Clipping: End Point Codes, Cohen Sutherland, Mid point Subdivision Algorithm(03), Mapping, dragging,echoing, Polygon Filling, character generation(03)
Lectures: 10

UNIT -III

3D graphics transformation-translation,rotation,scaling,reflection,shearing(02), Projection:parallel, Prespective projection(04), Hidden surface, removal algorithm method, back face removal algorithm,Z Buffer Algorithm, floating horizon technique(04)
Lectures: 10

UNIT -IV

Tweening-interpolation, morphing technique(02), GKS- primitive,work station(03), multimedia application-animation principle,animation tools(05)
Lectures: 10

Total Lectures:40

Text Books:
1. Mathematical Elements of Computer Graphics- Rogers(TMH)
2. Procedural Elements of Computer Graphics- Rogers(TMH)

Reference Book:
1. Computer Graphics- Hearn M. Baker (Pearson Education)

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER V

PAPER NUMBER: 506
PAPER NAME: Practical (C#)

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER VI

PAPER NUMBER:601
PAPER NAME: E-Commerce

CREDIT- I

Introduction to E-Commerce: Definition of E-Commerce, Scope of E-Commerce, driving forces for E-Commerce (03), Issues in implementing E-Commerce, E-Commerce Applications,Advantage- Business & Customer, Disadvantages (03), framework for understanding e- business, Classification of E-commerce, E-commerce Business Models (04).
Lectures: 10

CREDIT -II

Firewall: Types, Features, Characteristics(03), Electronic Data Interchange (EDI): Concept, Components, Differences between traditional EDI & Paper EDI, Advantages of EDI(04), Business Application of EDI, EDI Communication Process, EDI Security, Digital Signature(03).
Lectures: 10

CREDIT -III

Electronic Payment System: EPS Models, EPS Processing, Digital token based(02), debit card,smart card, Credit Card, risk in electronic payment system(03), E-auction: Introduction, Overview, Electronic trading(02), Online Banking: origin, advantages, disadvantages, Services(03).
Lectures: 10

CREDIT -IV

Web Security factors, E-Commerce security threats, security schemes, Protocols, Digital Certificates(03), Cyber law in India, Supply Chain Management (SCM): Components and issues(03), Customer Relationship Management (CRM): definition, Components, Benefits, ECRM: concept, impact, ECRM v/s CRM(04)
Lectures: 10

Total Lectures: 40

Text Book:

1. E-Commerce - Ritendra Goyal

Reference Books:

1. E-Commerce – Bharat Bhaskar
2. E-Commerce -- CSV Murthy

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER VI

PAPER NUMBER: 602
PAPER NAME: Linux & Shell Programming

UNIT - I

Linux introduction and file system – Basic Features, Advantage, Installing requirement , Basic Architecture of Unix/Linux System, Kernel , Shell(03).
Linux File System-Boot block, super block, Inode Table, data blocks, how Linux Access Files, storage files, Linux Standard directories(03). Commands for files and directories cd , ls, cp ,md, rm , mkdir , rmdi , pwd , file ,more, less, creating and viewing files using cat , file comparison- cmp & comm., view files, disk related commands, checking disk free spaces(04).
Lectures: 10

UNIT- II

Essential Linux commands, Understanding shells, Processes in linux- process fundamentals, connecting process with pipes, tee, Redirecting input output, manual help(03). Background processing ,managing multiple process, changing process priority with nice, scheduling of process at command, cron , batch commands ,kill ps, who, sleep, Printing commands find ,sort, touch , file ,file related commands ws, sat, cut , dd , etc.(03) Mathematical commands- bc , expr , factor , UNITs (02). Creating & editing files with vi, joe & vim editor (02).
Lectures: 10

UNIT- III

System Administration: Common Administrative tasks, identifying administrative files, configuration and log files, role of system administrator, managing user accounts- adding and deleting users, changing permission and ownership, creating and managing groups, modifying group attributes, temporary disable user accounts, creating and monitoringfile system(05) , checking and monitoring system performance, file security and permission, becoming super user using su. Getting system information with uname, host name, disk partition & sizes, users, kernel(03). Backup and restore files, reconfiguration hardware with kudzu, installing and removing packages with rpm command(02)
Lectures: 10

UNIT- IV

Shell programming- Basic of shell programming, various types of shells available in Linux, comparisons between various shells, shell programming in bash, read command, conditional and looping statements, case statements, parameter passing and arguments, Shell variables, system shell variables, shell keywords, Creating Shell programs for automate system tasks(05). Simple filter commands- pr, head, tail, cut, paste, sort, uniq, tr. Filter using regular expressions- grep, egrep and sed. Awk programming- report printing with awk(05).
Lectures: 10

Total Lectures- 40

Text & Reference books:

1. UNIX- Concepts & Applications (Third Ed.)- Sumitabha Das, TMH
2. Unix for programmers and users(Third Ed.) – Graham Glass & King Ables, Pearson Education India
3. Red Hat Linux 9 Bible- Christopher Negus , IDG Books India Ltd.
4. Stephen Prata- Advanced Unix – A Programmer's Guide, BPB Publication, 2008

BACHELOR OF COMPUTER APPLICATION(BCA)
SEMESTER VI

PAPER NUMBER: 603
PAPER NAME: Cryptography and Network Security

UNIT- I
Introduction to Network Security, Aspects of Network Security, Need For Security, Goals Of Network Security(04), Features of a good Security policy, security attacks, security services and mechanisms, network security,(02) Firewall: type, hardware and software, design principles(04)
Lectures: 10

UNIT- II
Encryption/decryption, conventional encryption model, conventional encryption(03) algorithms: character level encryption, bit-level encryption, private key encryption(02), Data encryption Standard algorithm, strength of DES(05)
Lectures: 10

UNIT- III
Public key encryption, principles of public key cryptography systems, RSA algorithm, security of RSA(03), Fermet's and euler's theorem, primality, The chineese remainder theorem(03), Digital signature, application for public key cryptosystems.**(**04)
Lectures: 10

UNIT- IV
E-mail security, PGP, S\MIME security, S\MIME functionality, cryptographic algorithms(03), IP Security: application of IPSec, Benefits of IPSec, IPSec architecture, IPSec Services, Authentication header, Encapsulating Security payload.(04) Web Security: threats on web, SSL and Transport layer Security. System security : virus and Intruders.**(**03)
.**Lectures: 10**

Total Lectures: 40

Text Books:
1. W. Stallings, Networks Security Essentials: Application & Standards, Pearson Education, 2000
2. W. Stallings, Cryptography and Network Security, Principles and Practice, Pearson

SEMESTER VI

PAPER NUMBER: 604
PAPER NAME: Industrial Training and Project